A Digital-Twin and Physical-Watermarking Testbed for Multi-Layer Intrusion Detection in SCADA Systems

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***Abstract*—Industrial supervisory control and data acquisition (SCADA) systems are increasingly targeted by attacks that falsify sensor telemetry or inject unauthorised control commands. Conventional network intrusion detection observes only the communication layer and is therefore blind to attacks that manipulate a field device directly. This paper presents a fully simulated, reproducible cyber-physical security testbed built around a liquid-metal boiler plant, and uses it to evaluate a defence that combines four complementary detection layers—network, process, an independent digital twin, and physical watermarking—fused by a correlation engine. The digital twin integrates its own physics model with datasheet parameters and a gated measurement-fusion rule, so a falsified reading produces a growing, detectable residual rather than silently corrupting the estimate. A bounded pseudo-random watermark is injected into an actuator command and its effect is checked against the twin-predicted response. Across ten reproducible attack scenarios, a network-only detector misses sensor-falsification attacks entirely (F1 = 0.00) while the full multi-layer stack detects them within 0.2 s (F1 = 0.94); conversely, command-path attacks are caught by the network layer, demonstrating that the layers are genuinely complementary. The platform runs live in a browser with a 10 Hz simulation, an operator HMI and an automated response engine, and is released with 56 automated tests.**

***Index Terms*—SCADA security, digital twin, intrusion detection, physical watermarking, cyber-physical systems, false data injection.**

I. INTRODUCTION

Modern industrial plants are controlled by SCADA systems in which programmable logic controllers (PLCs) drive actuators and read sensors, while a supervisory server polls those controllers over industrial protocols such as Modbus TCP, OPC UA and MQTT. The convergence of operational technology with commodity networking has widened the attack surface: incidents such as Stuxnet [1], the 2015–2016 attacks on the Ukrainian power grid [8], and the TRITON/TRISIS safety-system malware show that adversaries can falsify measurements, replay or inject control commands, and tamper with field devices.

A defender who watches only network traffic faces a fundamental blind spot. An attacker who compromises a field device, or who sits between a sensor and its controller, can present a plausible but false value to the operator while the physical process diverges. The network sees a well-formed packet; nothing in the communication layer is anomalous. Detecting such attacks requires reasoning about the physics of the process, not only its packets [4], [5].

This paper makes the following contributions: (1) a complete, reproducible cyber-physical security testbed—a simulated liquid-metal boiler plant with 37 sensors, 17 actuators, five PLCs/RTU and a Purdue-zoned control network—that runs live at 10 Hz with an operator HMI (Fig. 3); (2) an *independent* digital twin that fuses measurements through a gated correction, so that falsified telemetry is exposed rather than absorbed (Fig. 4); (3) a multi-layer IDS (network, process, twin, watermark) with a correlation engine that merges evidence into deduplicated, classified incidents (Fig. 6); (4) a functional physical-watermarking engine that injects a bounded excitation into the control path and verifies the measured process response; and (5) an attack laboratory of ten reproducible scenarios with ground-truth labelling, and a quantitative evaluation of the complementarity of the detection layers. The platform is an educational and research simulation: the process model is a simplified, plausible abstraction, not a validated engineering or safety model.

II. RELATED WORK

Process-aware intrusion detection. A large body of work augments network IDS with knowledge of the physical process, through invariant mining, state estimation and residual checks, or specification-based monitoring [4], [5]. Our process layer uses transparent, configuration-driven rules so that every alert carries human-readable evidence.

Digital twins for security. Digital twins have been proposed for anomaly detection by comparing predicted and observed behaviour [10]. A recurring risk is that a twin which is continuously corrected toward measurements will track a slow, stealthy falsification; we address this with a gated correction that keeps the twin anchored to physics.

Physical watermarking. Active detection by injecting a secret excitation was introduced to detect replay and false-data attacks that are stealthy to passive monitors [2], [3]. We integrate the result as one evidence source in a multi-layer IDS rather than as a stand-alone detector.

Testbeds. Physical testbeds such as SWaT and WADI [6], and network simulators such as MiniCPS [7], have enabled reproducible ICS-security research. Our contribution couples the process, network, twin, watermark, attack and response subsystems in one live, deterministic system.

III. SYSTEM ARCHITECTURE

The platform is a Python (FastAPI) backend and a React/TypeScript frontend. The backend advances a single simulation engine at a fixed timestep (dt = 0.1 s, i.e. 10 Hz) and streams compact state snapshots to the browser over a WebSocket; a REST API exposes control endpoints, and SQLite persists alerts, incidents, events, responses and experiment results. The system is organised as the layered stack of Fig. 1.

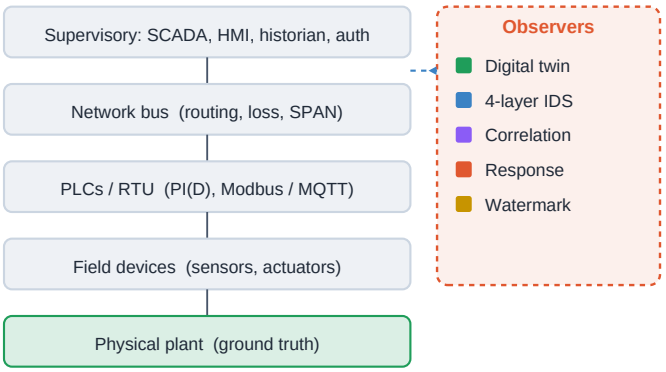


Fig. 1. Layered architecture. The attacker may touch only the network and field devices; the twin, watermark, IDS and response engine observe the stack.

Each 10 Hz tick executes the fixed sequence of Fig. 2. Four design invariants, enforced by automated tests, make the evaluation meaningful: (i) the attacker can touch only the network and field-device firmware hooks; (ii) the twin never reads ground-truth state; (iii) the safety system is independent of the network and the response engine; and (iv) ground-truth attack labels reach the evaluator only, never the detectors.

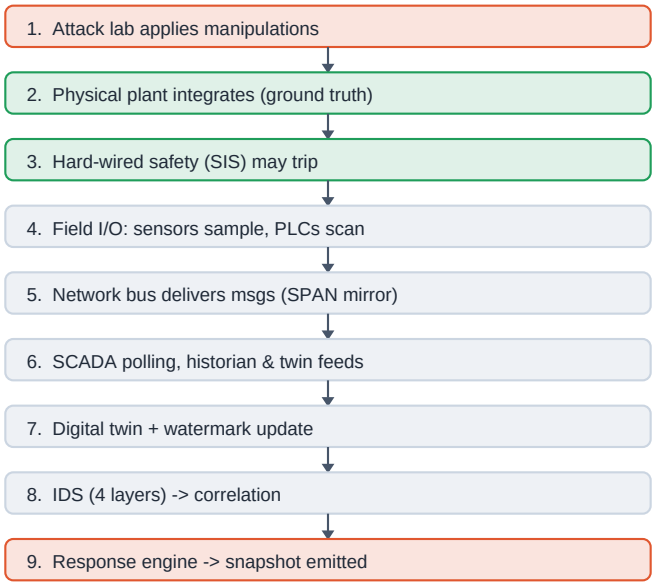


Fig. 2. Per-tick data-flow of the simulation engine (10 Hz). Green steps read the true plant state; the attacker and the SPAN-fed IDS never see it.

IV. PROCESS AND NETWORK SIMULATION

Process model. The primary plant is a lumped-parameter liquid-metal boiler with an intermediate heat exchanger (IHX), a secondary water loop and a turbine-generator (Fig. 3). The bulk vessel temperature evolves as a heat balance:

$$C_b\, dT_b/dt = Q_{heater} - \dot{m} c_p\,(T_b - T_{cl}) - UA_{loss}\,(T_b - T_{amb}) \quad (1)$$

The hot leg is a transport lag of the bulk temperature, the IHX uses a counter-flow effectiveness–NTU relation, and the secondary loop and generator follow first-order dynamics. Four auxiliary subsystems reuse the same abstractions: an induction motor, a water-treatment plant, a batch chemical-mixing reactor, and a pipeline/tank farm. Parameters are configurable and explicitly hypothetical.

Field devices. Each sensor turns the true value into a measurement via additive noise, quantisation and range clamping; each actuator applies a rate limit and lag. Firmware hooks on these devices are the only point at which the attack laboratory can tamper with a field element.

Control and safety. Five controllers (four Modbus PLCs and one MQTT RTU) run PI(D) loops and sequencers. A separate Safety Instrumented System reads the plant through its own instruments, is not on the network, and trips final elements *after* any firmware tampering—so an emergency shutdown cannot be defeated by a compromised device.

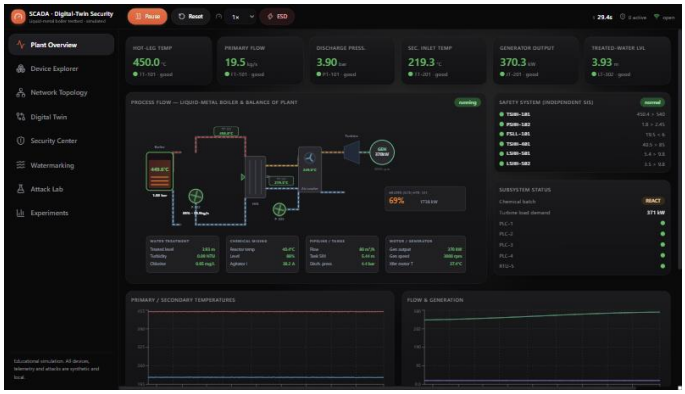


Fig. 3. Live operator HMI: the animated P&ID of the liquid-metal boiler, balance-of-plant tiles, independent safety panel and live trend charts.

Network and protocols. The network is a Purdue-zoned graph (Fig. 5). Modbus TCP is encoded and decoded as spec-conformant application data units (MBAP header plus PDU; function codes 03/04/06/16 and exceptions) [9]; OPC UA and MQTT are modelled at the application level and labelled as such. A discrete-event bus carries every message with latency, jitter and loss, and mirrors supervisory traffic to the IDS through a SPAN port.

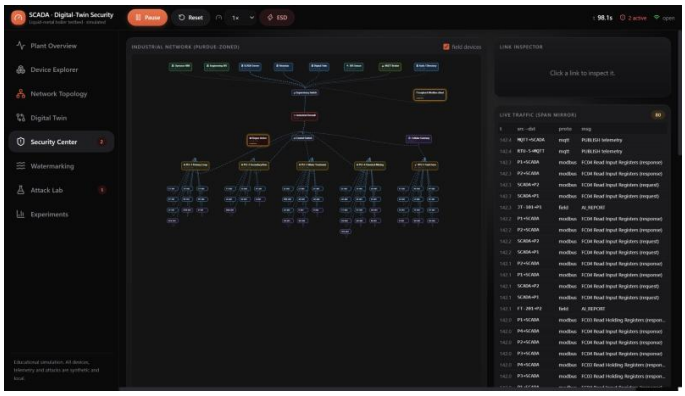


Fig. 5. Purdue-zoned network: supervisory LAN, firewall, control LAN with PLCs and field devices, a cellular gateway to the RTU, and the live SPAN-mirrored traffic.

V. INDEPENDENT DIGITAL TWIN

The digital twin integrates its own copy of the primary-loop physics with a fourth-order Runge–Kutta scheme and datasheet parameters that differ from the plant by a few percent. Its inputs are the issued commands (plus the trusted watermark offset) and a few measured drivers; it never reads the plant's hidden ground-truth state. Each tick the twin forms a one-step-ahead prediction and applies a *gated correction*. The residual and normalised innovation are:

$$r_k = y_k - \hat{y}_k, \quad z_k = (r_k - b_k)/\sigma_k \quad (2)$$

A measurement updates the state only if $|z_k| < 5$; readings that jump outside the gate are rejected. A falsified sensor therefore cannot silently drag the twin along, so a stealthy lie produces a persistent, growing residual (Fig. 4). Residuals feed a two-sided CUSUM statistic for persistence and a counter for sudden errors, with a calibration window that estimates each residual's bias and spread.

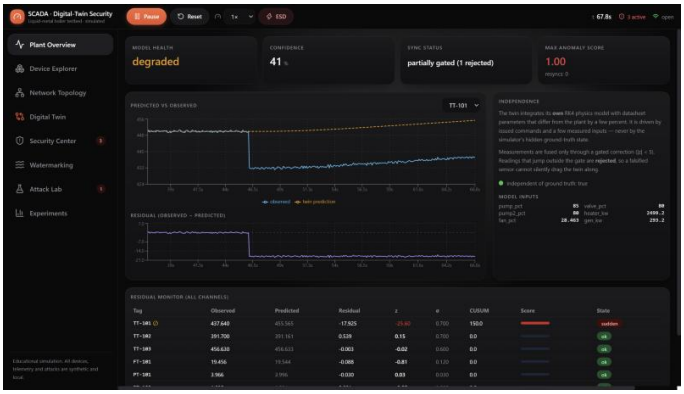


Fig. 4. Twin view during a false-data injection: the observed value (blue) is pulled to the falsified reading while the twin prediction (orange) tracks the real temperature; the residual reaches z = −25.6.

VI. MULTI-LAYER INTRUSION DETECTION

Detection is organised as four independent layers. The **network** layer (SPAN mirror only) flags rogue devices, spoofing, unexpected flows, floods, invalid sequences and replays, and unauthorised or out-of-envelope writes. The **process** layer (SCADA observations) flags limit violations, frozen signals, implausible rate-of-change, setpoint deviation, pump-flow inconsistency and energy-balance violations. The **digital-twin** layer flags persistent and sudden residuals and correlated multi-sensor anomalies. The **watermark** layer contributes its consistent / suspicious / inconclusive decision.

Each detector emits conditions; a finding manager deduplicates them into alert records. A correlation engine groups findings in the same zone and time window into incidents and combines per-layer confidence as

$$C = 1 - \prod_i (1 - c_i) \quad (3)$$

raising severity by one level per corroborating layer, classifying the incident with transparent rules, and recommending responses—without double-counting duplicate alerts (Fig. 6).

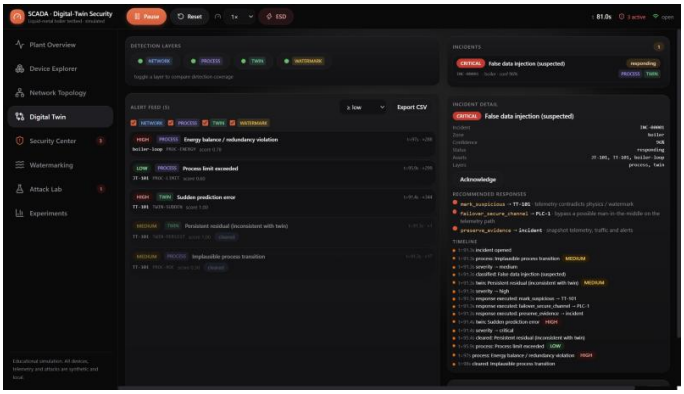


Fig. 6. IDS Security Center during a false-data injection: the alert feed, the correlated critical incident at 96% confidence, and the automatically executed responses.

VII. PHYSICAL WATERMARKING

The watermark engine injects a reproducible, bounded pseudo-random binary sequence (PRBS) into an eligible actuator command inside the PLC's secured function block; the exported command register still reports the operator's base value. The excitation for slot k is a pure function of a seed and k, so the injector and the detector generate identical sequences without exchanging them. Two shadow copies of the twin's hydraulics compute the *expected* response; high-pass-filtered telemetry gives the *observed* response. For each channel the detector computes the correlation ρ, the response gain and an SNR, and returns *consistent*, *suspicious*, or *inconclusive*. Injection self-suspends on a safety trip, near saturation, close to a process limit, or when measurement quality is poor. In baseline operation ρ ≈ 0.80 (flow) and ≈ 0.95 (pressure); a replay that smooths away the signature is reported as suspicious.

VIII. ATTACK LABORATORY AND RESPONSE

Ten reproducible scenarios exercise the detectors: false sensor-data injection, frozen readings, actuator-response manipulation, unauthorised setpoint writes, command replay, supervisory-link denial of service, insider writes, digital-twin model tampering, watermark interference, and a coordinated process-plus-network attack. Each records its seed and parameters, produces real changes, exposes a ground-truth label, and supports stop/recover. During telemetry falsification the physical process keeps evolving from real actuator action while only the *observed* values are altered—the gap the twin, redundancy and watermark expose. A response engine operates in monitor, operator-approved or automatic mode, acting through simulated effects (quarantine, write allow-list, setpoint restore, account disable, secure-channel failover, safe state, twin resync, evidence preservation); it is kept separate from the safety interlocks.

IX. METHODOLOGY AND RESULTS

Experiments run a deterministic engine (seed 1337) through baseline, attack and recovery phases, recording a per-sample trace of labels, active findings, setpoint deviation and incident state. The *same* trace is scored under four detector configurations, so the comparison is apples-to-apples. Each 2 s window is positive if the attack was active for at least half of it, and detected if a non-info finding mapped to the attacked zone. Ground-truth labels feed only the evaluator; seeded streams make results replay exactly. Table I reports detection F1; Fig. 7 isolates the two telemetry-falsification attacks.

TABLE I. Detection F1 by detector configuration (seed 1337, 2 s windows). "miss" = no detection.

Scenario	Net only	Net+ Proc	+Twin	Full +WM	Lat.
A. Normal operation	–	–	–	–	–
B. False data injection	miss	0.94	0.94	0.94	0.2s
C. Watermark interference	miss	miss	0.71	0.96	1.1s
D. Unauthorised setpoint	0.91	0.91	0.91	0.91	0.0s
E. Command replay	0.93	0.93	0.93	0.93	0.5s
F. Coordinated	0.91	0.91	0.91	0.91	0.5s

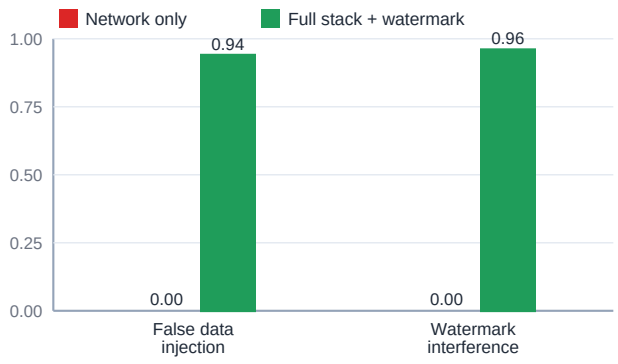


Fig. 7. On telemetry-falsification attacks a network-only IDS scores F1 = 0, while the full multi-layer stack reaches 0.94–0.96.

Four observations follow. (1) Normal operation raises no sustained alerts. (2) Network-only is blind to telemetry falsification (B, C): these

produce no network evidence; the process and twin layers recover B within 0.2 s. (3) The watermark earns its place on C: the process layer sees nothing, the twin partially catches the drift (0.71), and the watermark raises detection to 0.96. (4) Command-path attacks (D, E, F) leave direct network evidence and are caught even by network-only. In a live run of B, the real hot-leg temperature rose to 459.8 °C while SCADA showed the falsified 442.3 °C; the twin residual reached $z = -24.9$, the engine raised a critical incident at 0.96 confidence, and the automatic response quarantined the source, restored the setpoint and preserved evidence.

X. DISCUSSION AND LIMITATIONS

The detection layers are complementary, not redundant: command-path attacks are visible on the network, telemetry-falsification attacks are visible to process/twin detection, and an erased signature is caught by the active watermark. A defender needs all four, fused by correlation. The process model is a simplified, hypothetical abstraction, not a validated engineering model. OPC UA and MQTT are emulated at the application level. The twin shares a model family with the plant, so a slow, in-gate ramp can still bias it; the watermark and redundancy checks mitigate this, and quantifying that boundary is future work. The evaluation uses a single seed per scenario.

XI. CONCLUSION

We presented a self-contained, reproducible cyber-physical security testbed coupling a simulated SCADA plant with an independent digital twin, a four-layer IDS, a physical-watermarking engine, an attack laboratory and an automated response engine, running live at 10 Hz with an operator HMI. We showed quantitatively that network-only detection is blind to telemetry falsification while the multi-layer stack detects it within fractions of a second, and that the physical watermark adds value where passive layers fail. Future work includes characterising the stealthy in-gate ramp boundary, adding learning-based detectors, broadening the stochastic evaluation, and binding real protocol libraries over an isolated loopback network.

REFERENCES

[1] N. Falliere, L. O. Murchu, and E. Chien, “W32.Stuxnet Dossier,” Symantec, 2011.

[2] Y. Mo and B. Sinopoli, “Secure control against replay attacks,” in Proc. 47th Allerton Conf., 2009, pp. 911–918.

[3] Y. Mo, S. Weerakkody, and B. Sinopoli, “Physical authentication of control systems,” IEEE Control Syst. Mag., vol. 35, no. 1, pp. 93–109, 2015.

[4] A. A. Cárdenas et al., “Attacks against process control systems,” in Proc. ACM ASIACCS, 2011, pp. 355–366.

[5] D. Urbina et al., “Limiting the impact of stealthy attacks on ICS,” in Proc. ACM CCS, 2016, pp. 1092–1105.

[6] J. Goh et al., “A dataset to support research in secure water treatment systems,” in Proc. CRITIS, 2016.

[7] D. Antonioli and N. O. Tippenhauer, “MiniCPS: A toolkit for security research on CPS networks,” in Proc. ACM CPS-SPC, 2015, pp. 91–100.

[8] R. M. Lee, M. J. Assante, and T. Conway, “Analysis of the cyber attack on the Ukrainian power grid,” E-ISAC/SANS, 2016.

[9] Modbus Organization, “Modbus Application Protocol Specification V1.1b3,” 2012.

[10] M. Eckhart and A. Ekelhart, “Towards security-aware virtual environments for digital twins,” in Proc. ACM CPSS, 2018, pp. 61–72.